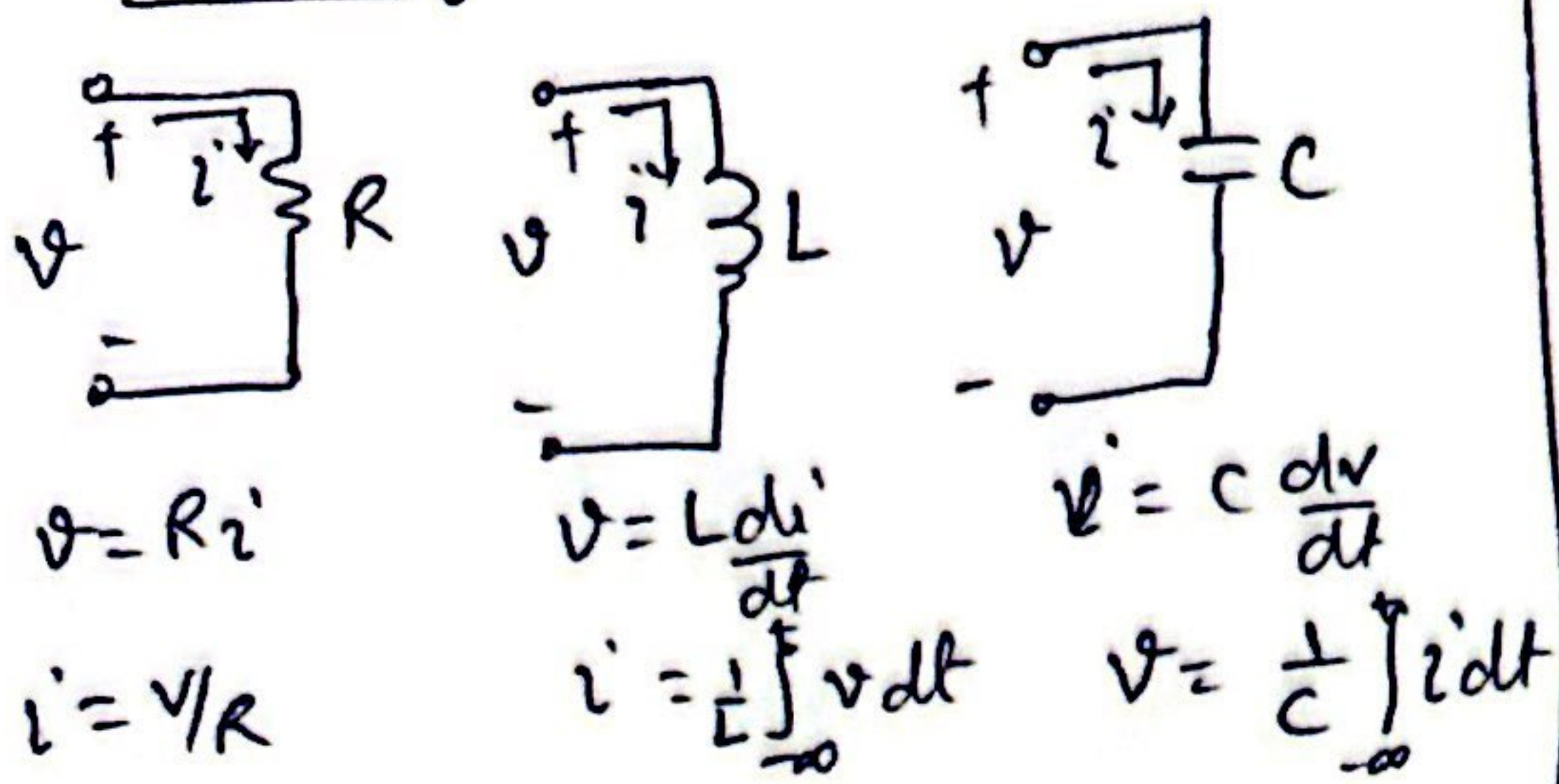
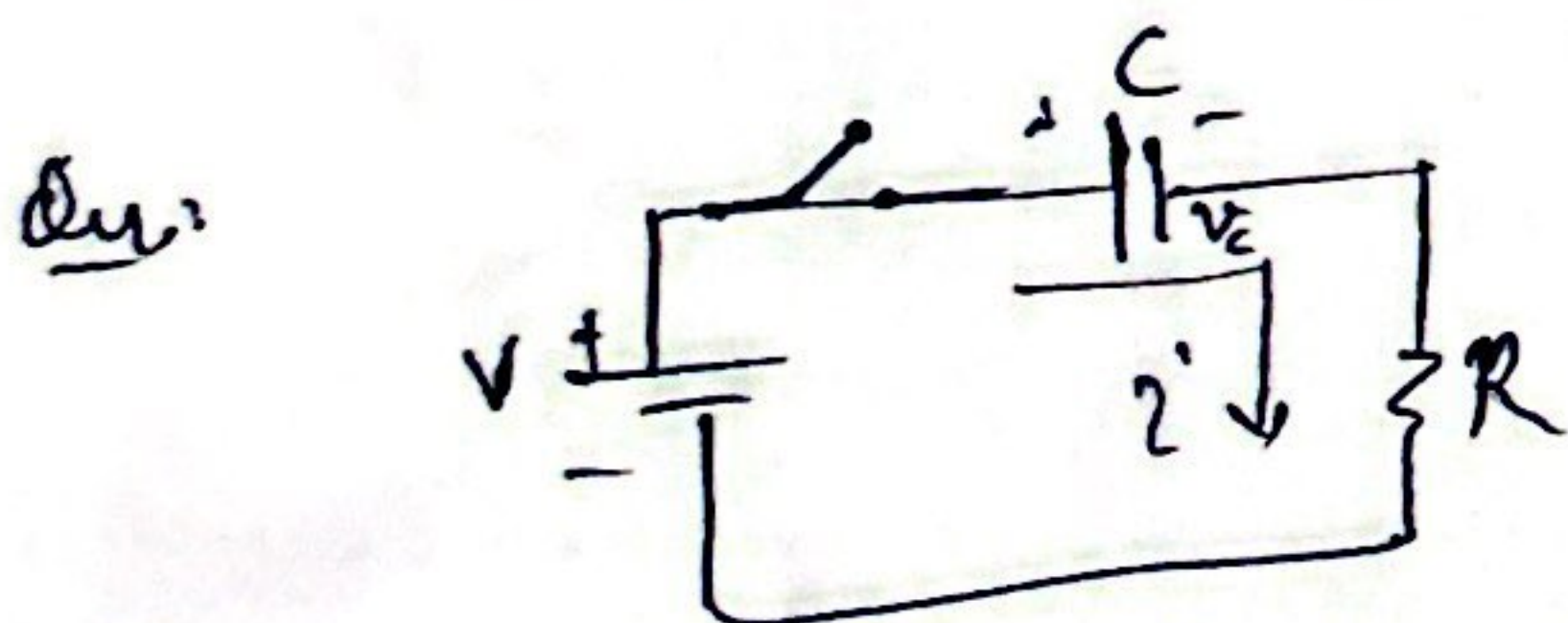
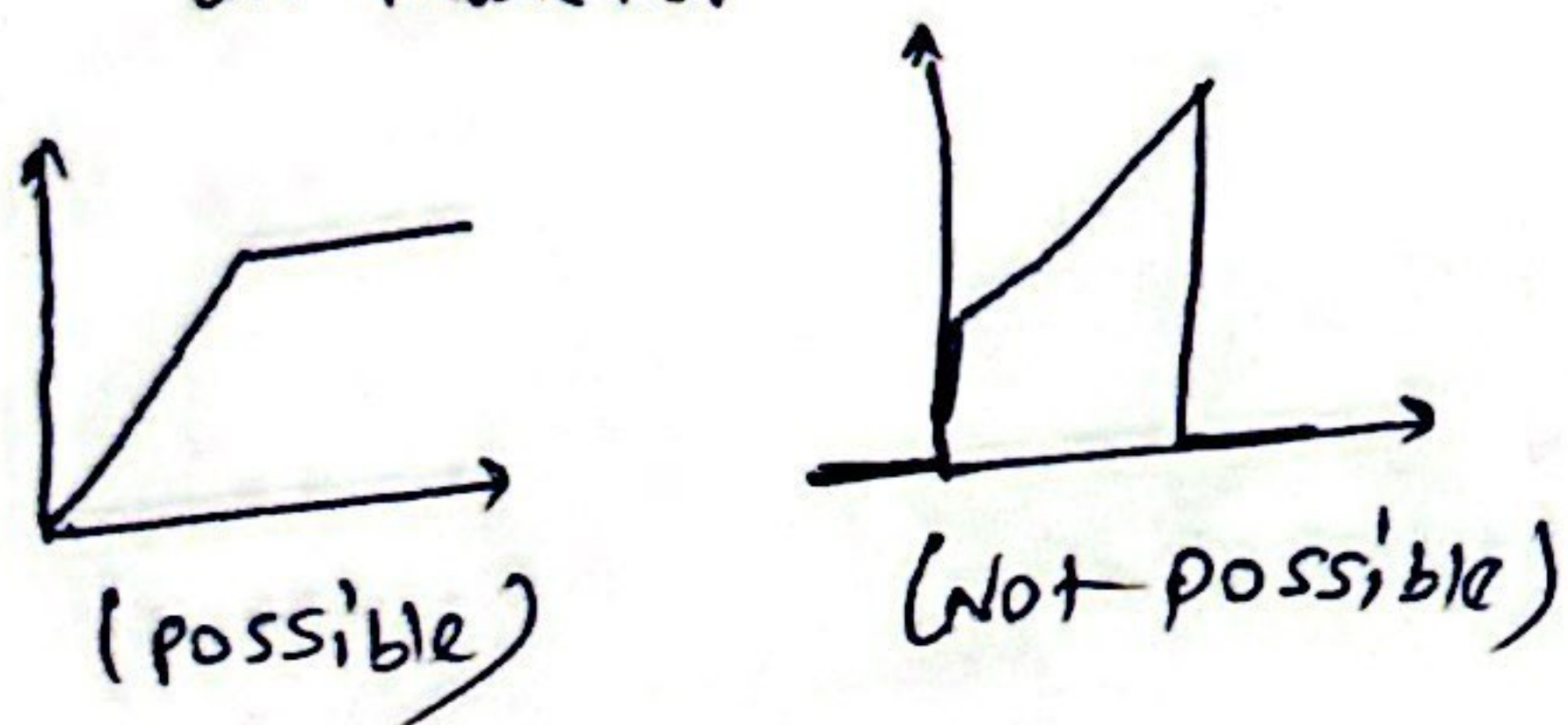


③ Summary:



⇒ Voltage across capacitor & current in an inductor can not change instantaneously.

Ques: which can be current through an inductor

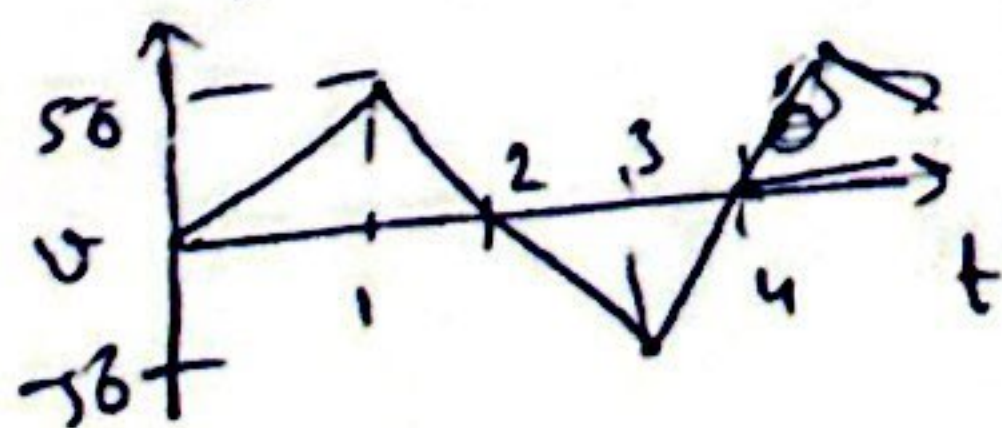


if switch closes at $t = 0$ & cap is initially relaxed, initial condition $i(0^+)$, $v_c(0^+)$, $v_c'(0^+)$

$$v_c(0^+) = 0 \Rightarrow i(0^+) = \frac{V}{R}$$

$$\frac{dv_c(0^+)}{dt} = i/c = \frac{V}{RC}$$

Ques: determine the current as fn. of time through a 200 mF whose voltage is



$$v = \begin{cases} 50t & 0 < t < 1 \\ 100 - 50t & 1 < t < 3 \\ -200 + 50t & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

$$i = C \frac{dv}{dt} = \begin{cases} 10 \text{ mA} & 0 < t < 1 \\ -10 \text{ mA} & 1 < t < 3 \\ 10 \text{ mA} & 3 < t < 4 \\ 0 \text{ mA} & \text{otherwise} \end{cases}$$

Linearity: homogeneity & Additivity

1) homogeneity property

$$\text{if } x(t) \rightarrow y(t)$$

$$\text{then } \alpha x(t) \rightarrow \alpha y(t)$$

2) Additive property

$$\text{if } x_1(t) \rightarrow y_1(t) \text{ and}$$

$$x_2(t) \rightarrow y_2(t)$$

$$\text{then } x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$$

Check if R, L, C are linear

KCL - KVL

analyzing circuits

Ex. $v = 1V$
 $i = \frac{1}{1} = 1A$

Ex. $i = 1A$
 $v = i \times R = 1V$

Ques: $i = ?$

A) 1A B) 2A C) 0A D) can not determine

⇒ since voltage generated by current source is same as voltage source, voltage source is not providing any current.

⇒ if current source is 2A, then 1A current flows into the voltage source.

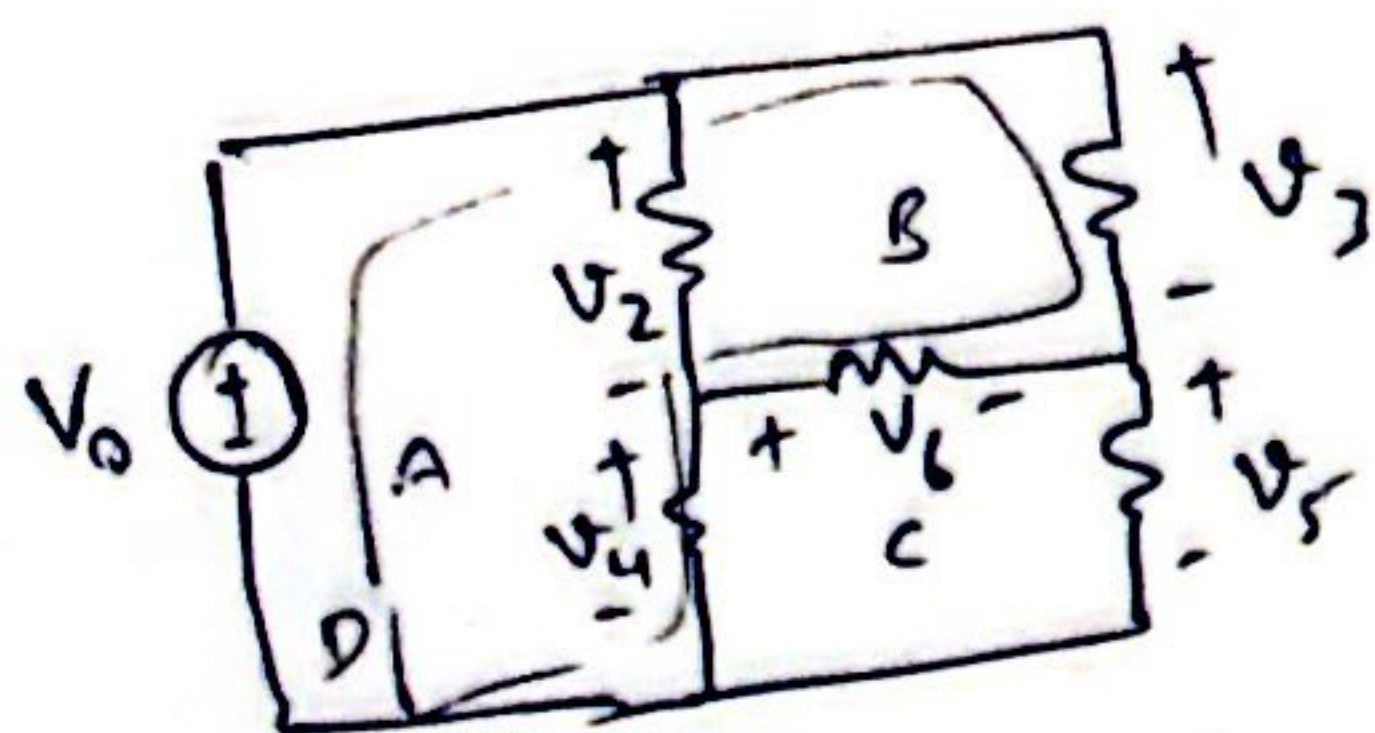
More complex ckt & can be analyzed using KVL & KCL

KVL: Algebraic sum of all voltages around a closed path is zero

$$-V_s + V_{R1} + V_{R2} = 0$$

$$-V_s + iR_1 + iR_2 = 0$$

4)



how many loops? total 7

$$A: -V_0 + v_2 + v_4 = 0$$

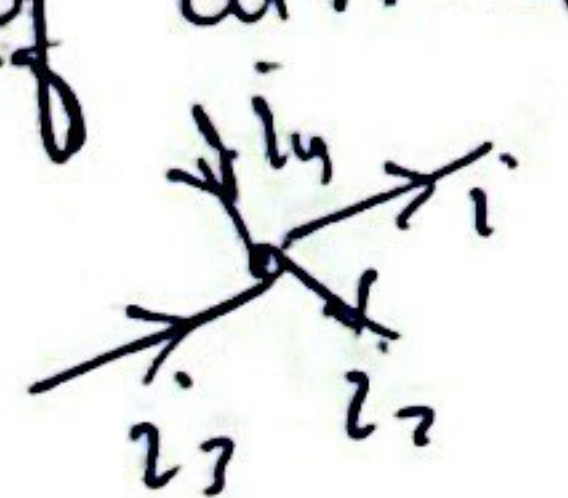
$$B: -v_2 + v_3 - v_6 = 0$$

$$C: -v_4 + v_6 + v_5 = 0$$

$$D: -V_0 + v_3 - v_6 + v_4 = 0$$

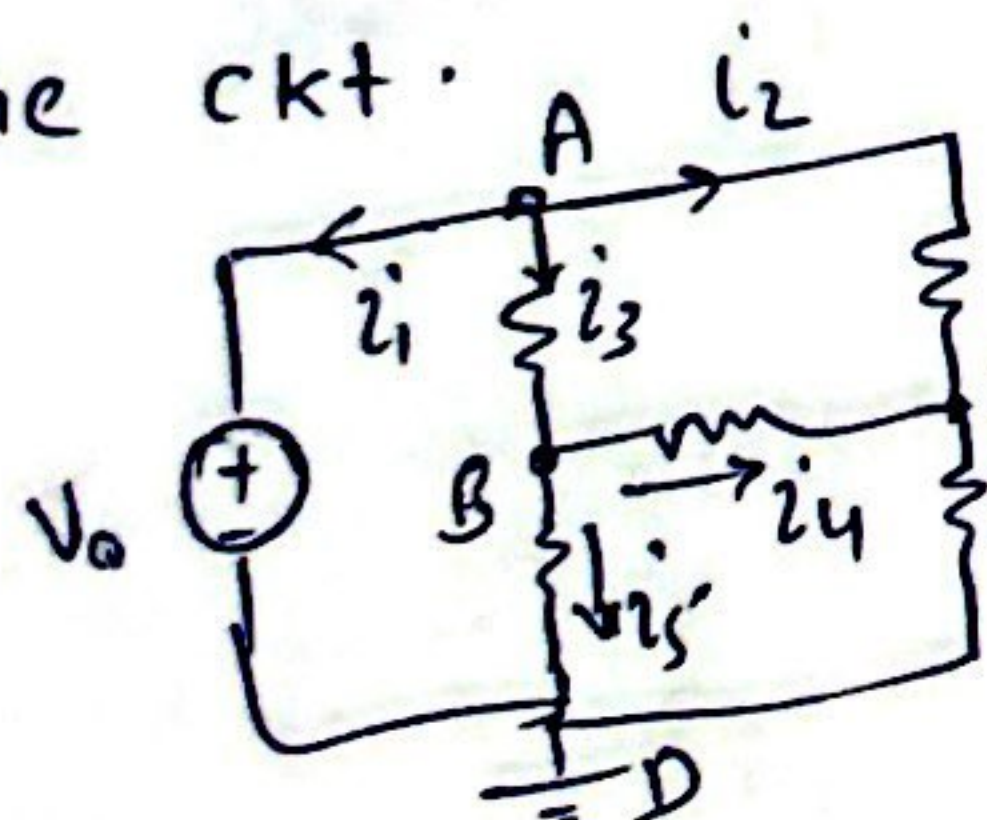
7 eqⁿs can be written but not all are lin. independent

KCL: at any current entering into one node is equal to sum of current leaving



$$i_1 + i_2 - i_3 - i_4 = 0$$

Same ckt.



find nodes

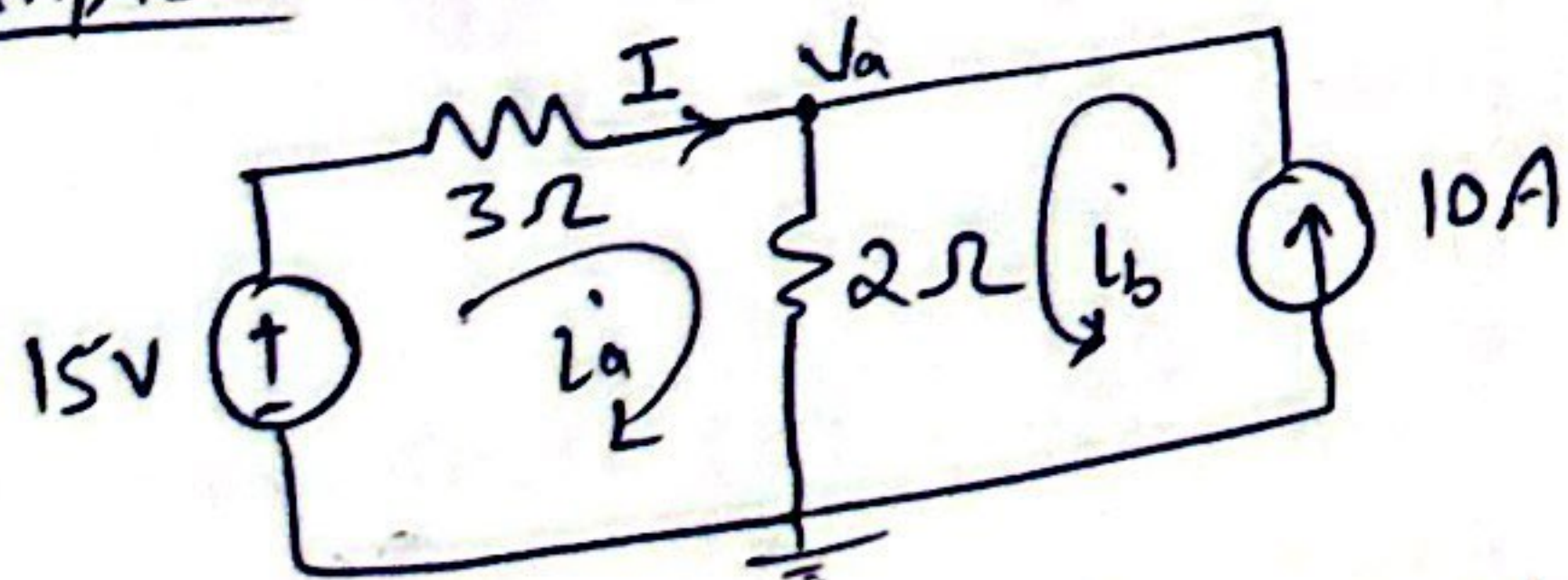
$$A: i_1 + i_2 + i_3 = 0$$

$$B: i_4 + i_5 - i_3 = 0$$

$$C: i_6 - i_4 - i_2 = 0$$

$$D: -i_6 - i_5 - i_1 = 0$$

Examples:



$$\text{Node method: } \frac{V_a - 15}{3} - 10 + \frac{V_a}{2} = 0$$

$$\Rightarrow V_a = 18 \text{ volts.}$$

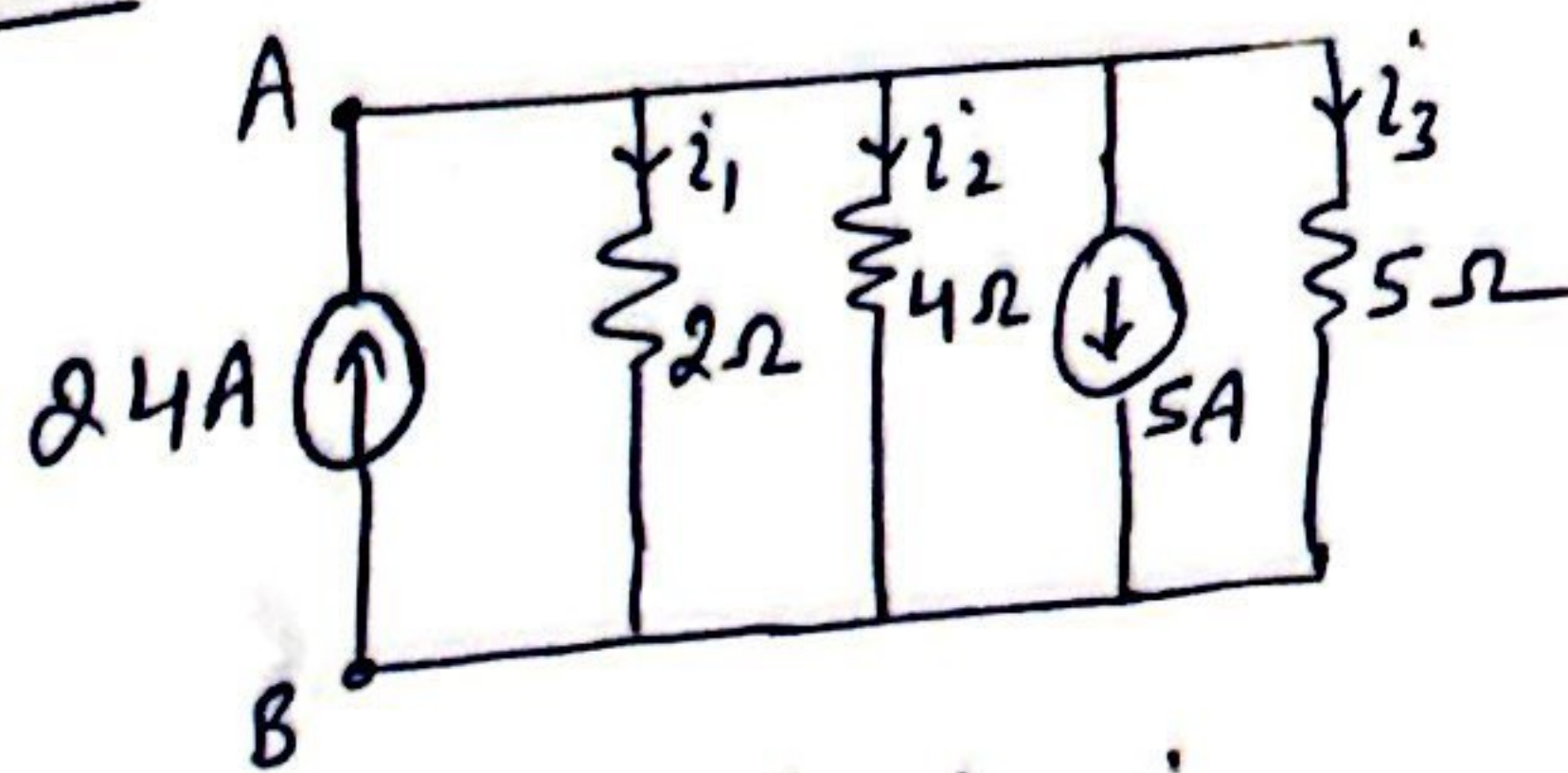
$$\Rightarrow I = (15 - 18)/3 = -1 \text{ A}$$

loop method:

$$a: -15 + 3i_a + 2(i_a + i_b) = 0$$

$$b: i_b = 10 \Rightarrow i_a = -1 \text{ A}$$

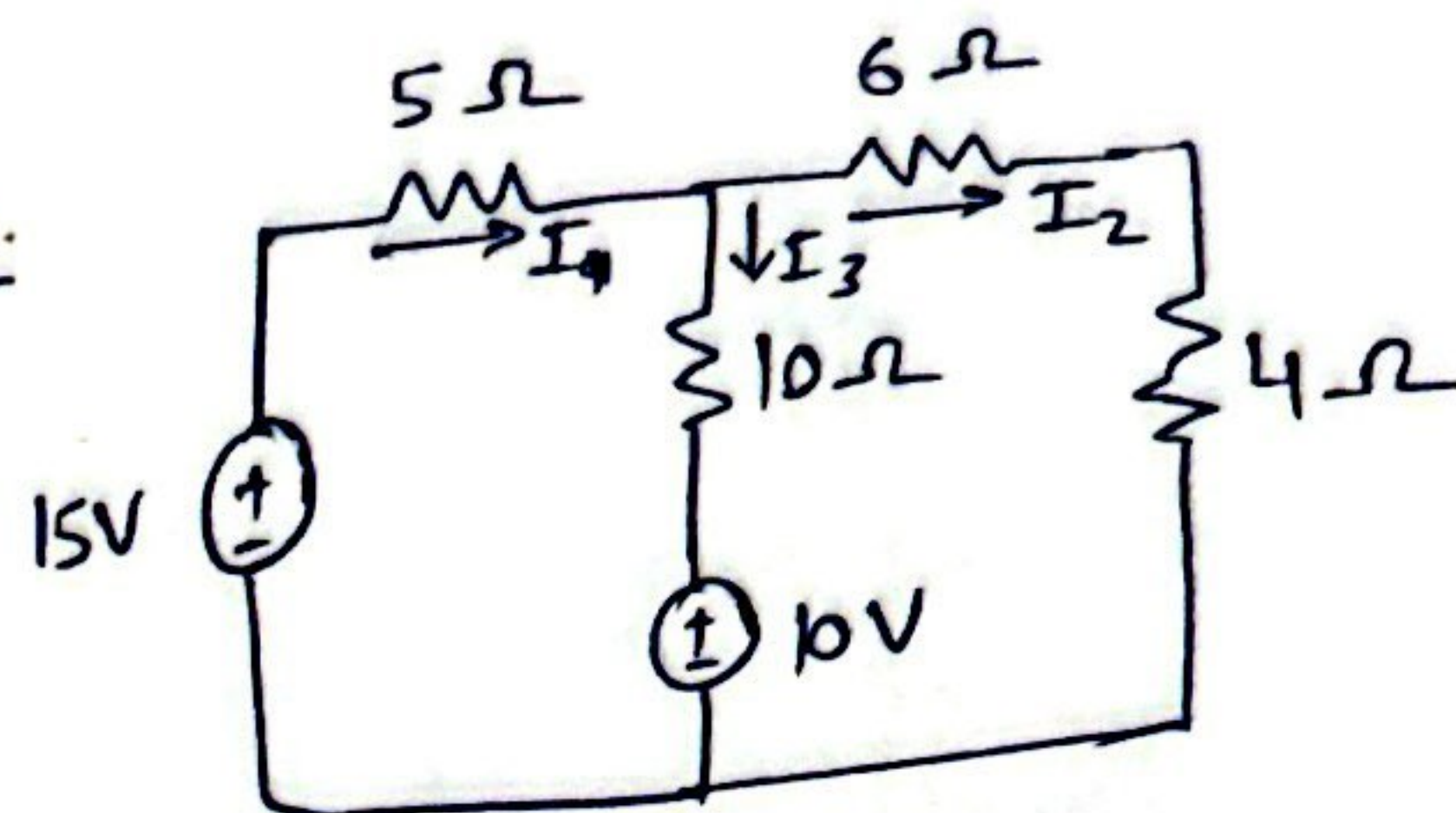
Ex:



find V_{AB} , i_1 , i_2 , i_3

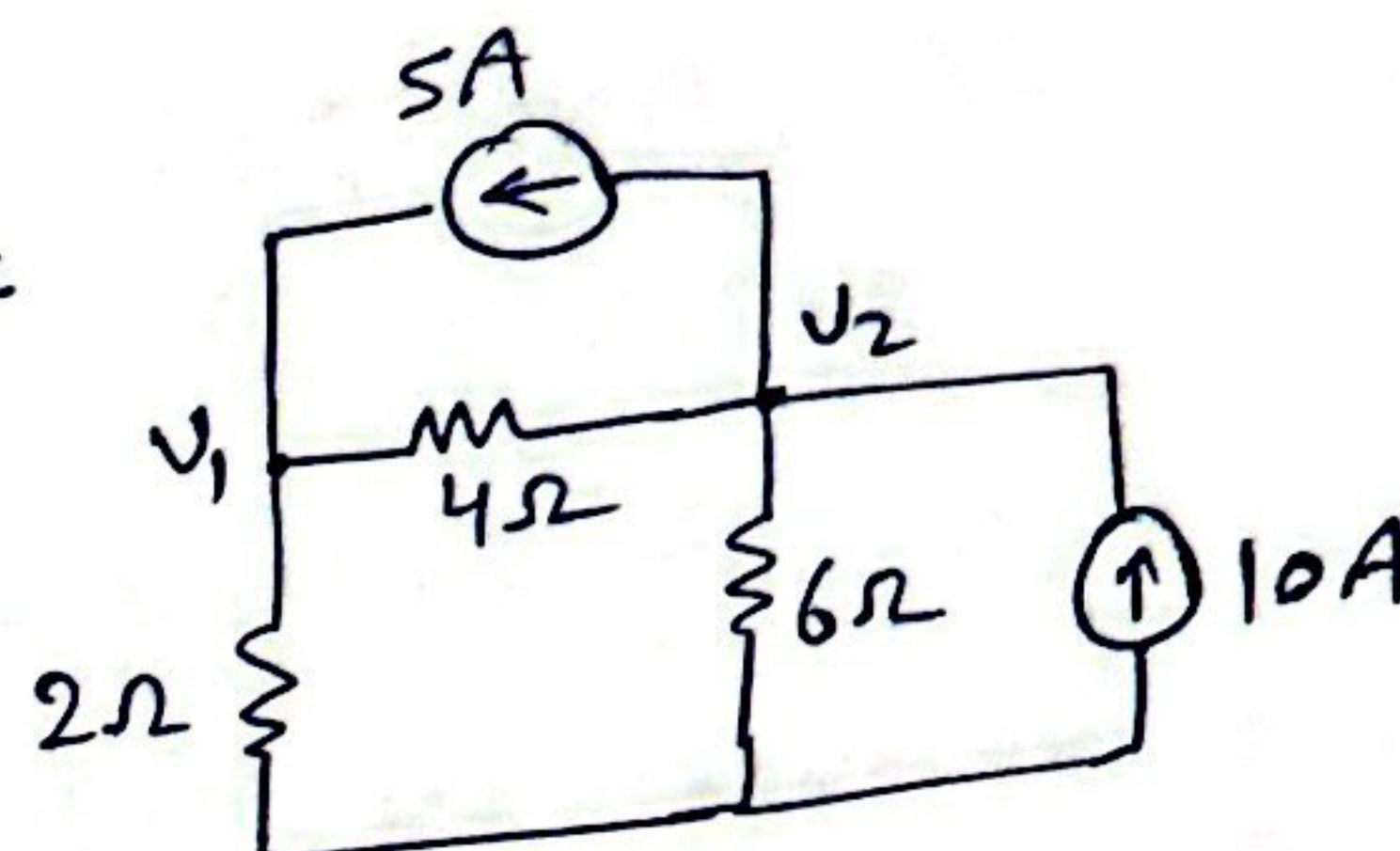
$$\text{Ans: } V_{AB} = 20 \text{ V, } i_1 = 10, i_2 = 5, i_3 = 4$$

Ex:



$$\text{Ans: } I_1 = 1 \text{ A, } I_2 = 1 \text{ A, } I_3 = 0$$

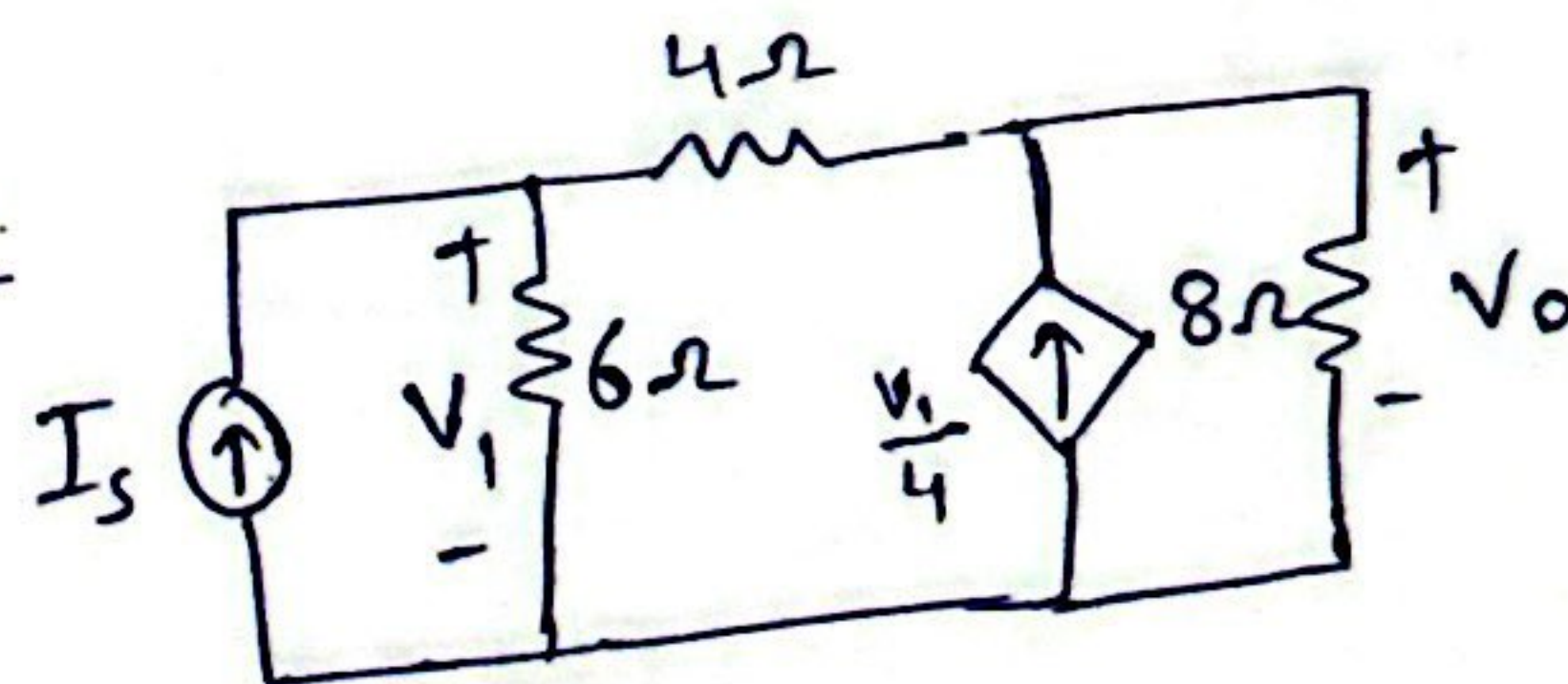
Ex:



find V_1 , V_2 .

$$\text{Soln: } V_1 = 13.33 \text{ V ; } V_2 = 20 \text{ V}$$

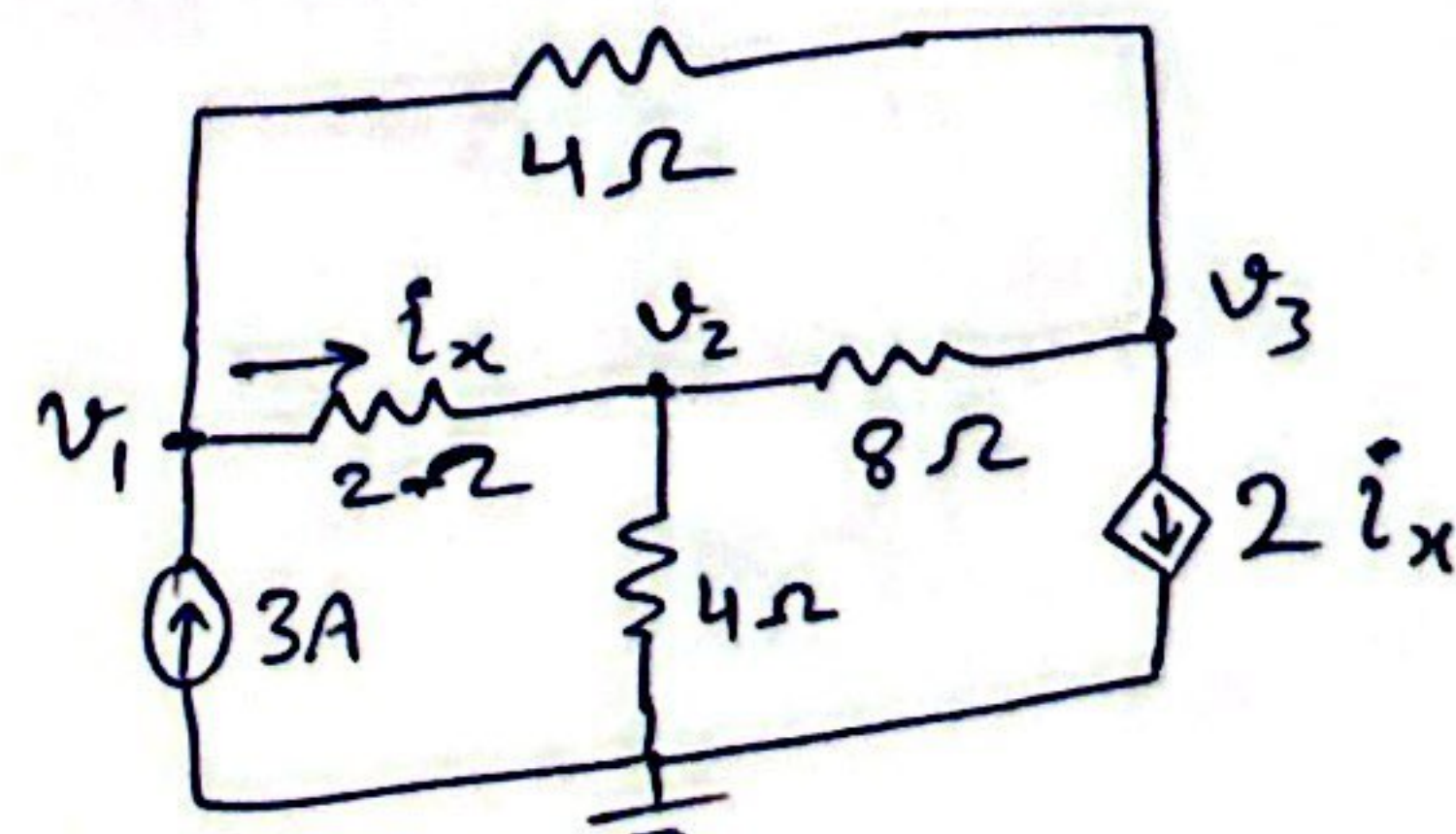
Ex:



find I_s , if $V_0 = 16 \text{ V}$

$$\text{Ans: } I_s = 1 \text{ A}$$

Ex:

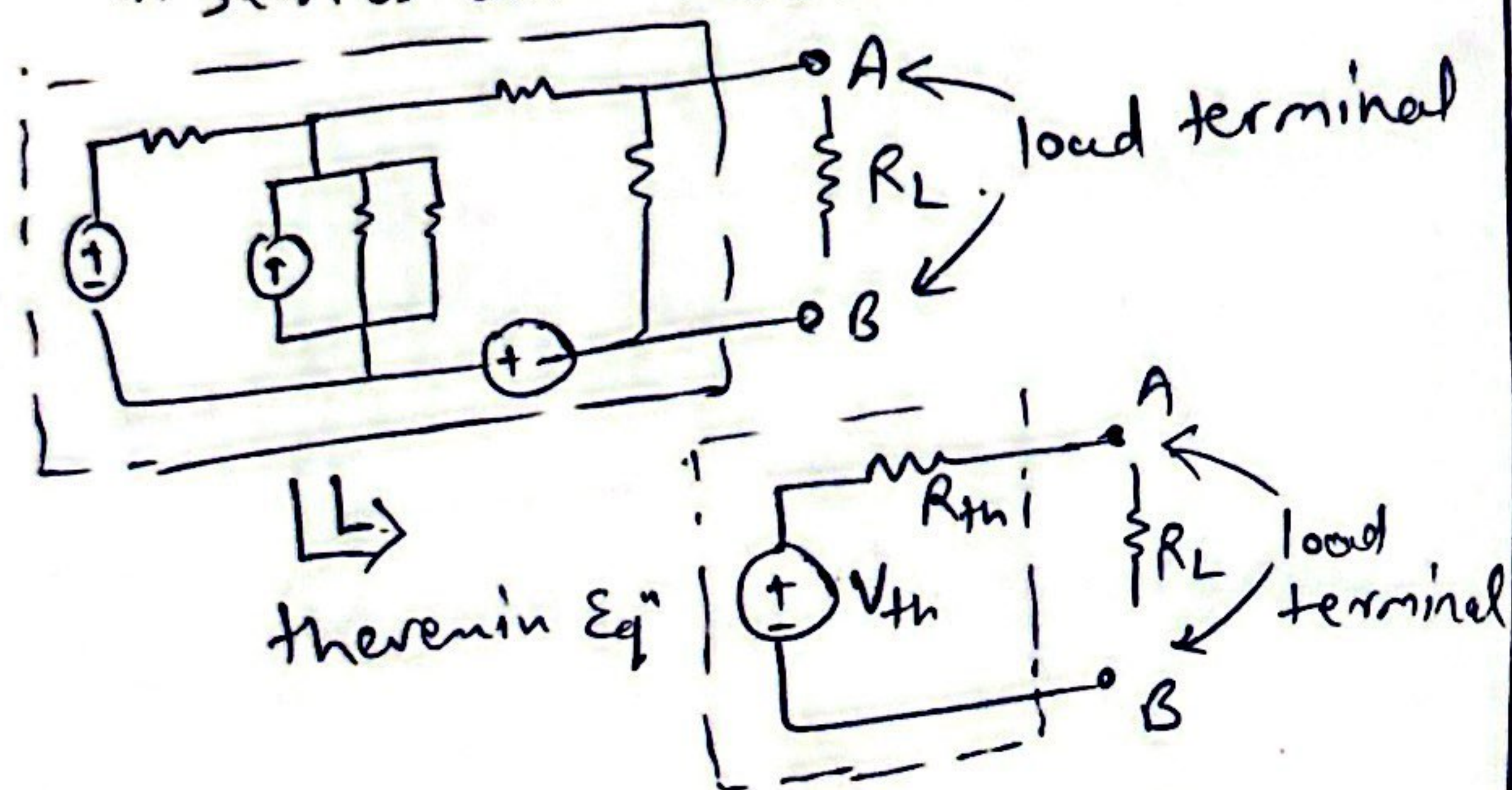


$$\text{Ans: } V_1 = 4.8, V_2 = 2.4, V_3 = -2.4$$

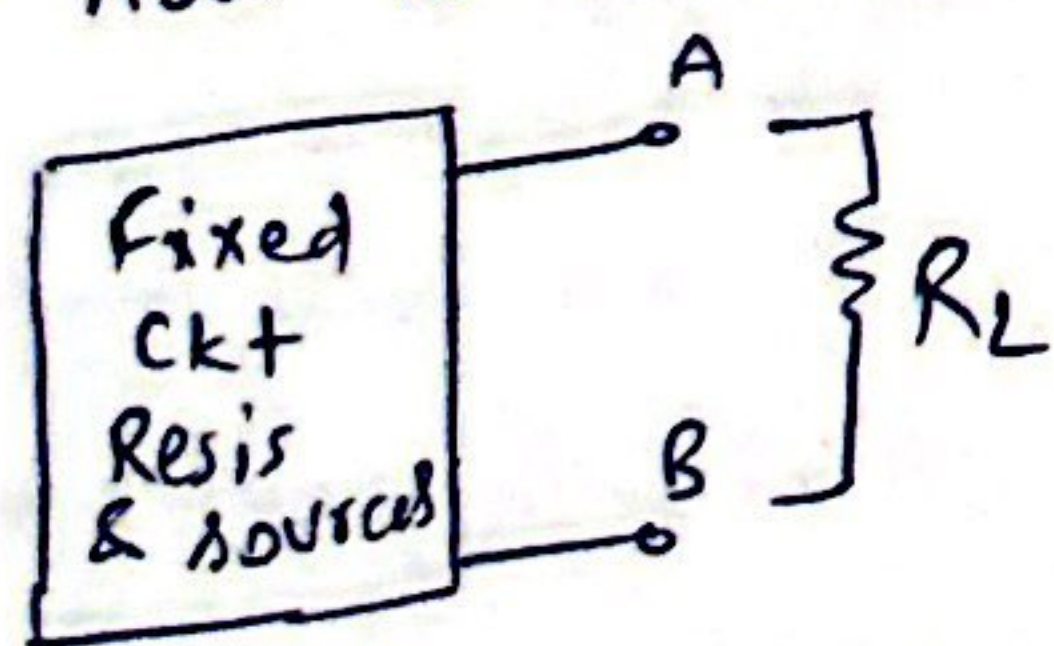
5

Thevenin's Theorem

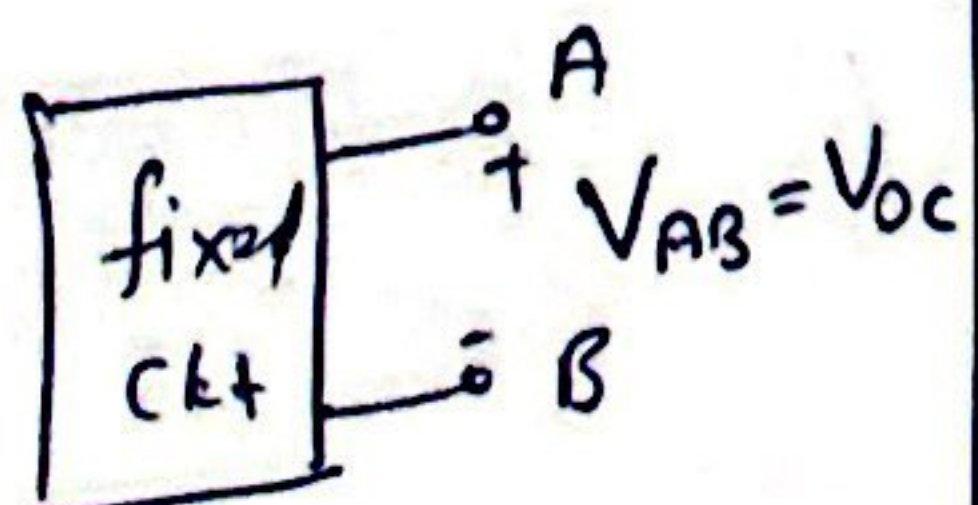
- it often occurs that a particular element in the ckt is variable (usually 'load') while other elements are fixed.
- Ex: home outlet terminal is connected to different appliances
- every time 'load' is changed, ckt needs to be analyzed all over again
- To avoid this problem, thevenin theorem provides a technique through which fixed part of ckt can be replaced by a voltage source in series with resistance.



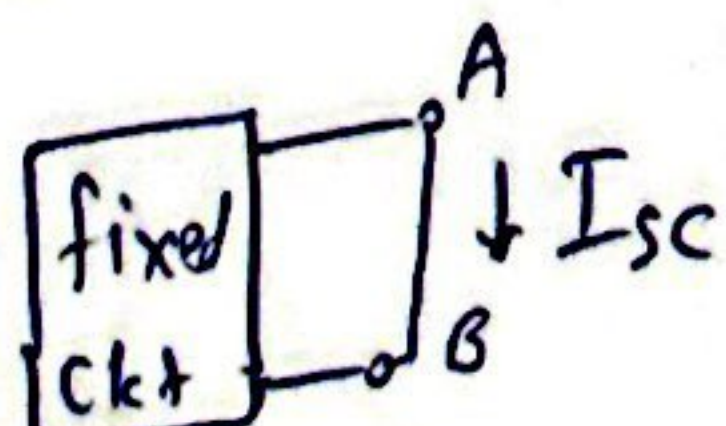
⇒ how to determine V_{th} & R_{th} ?



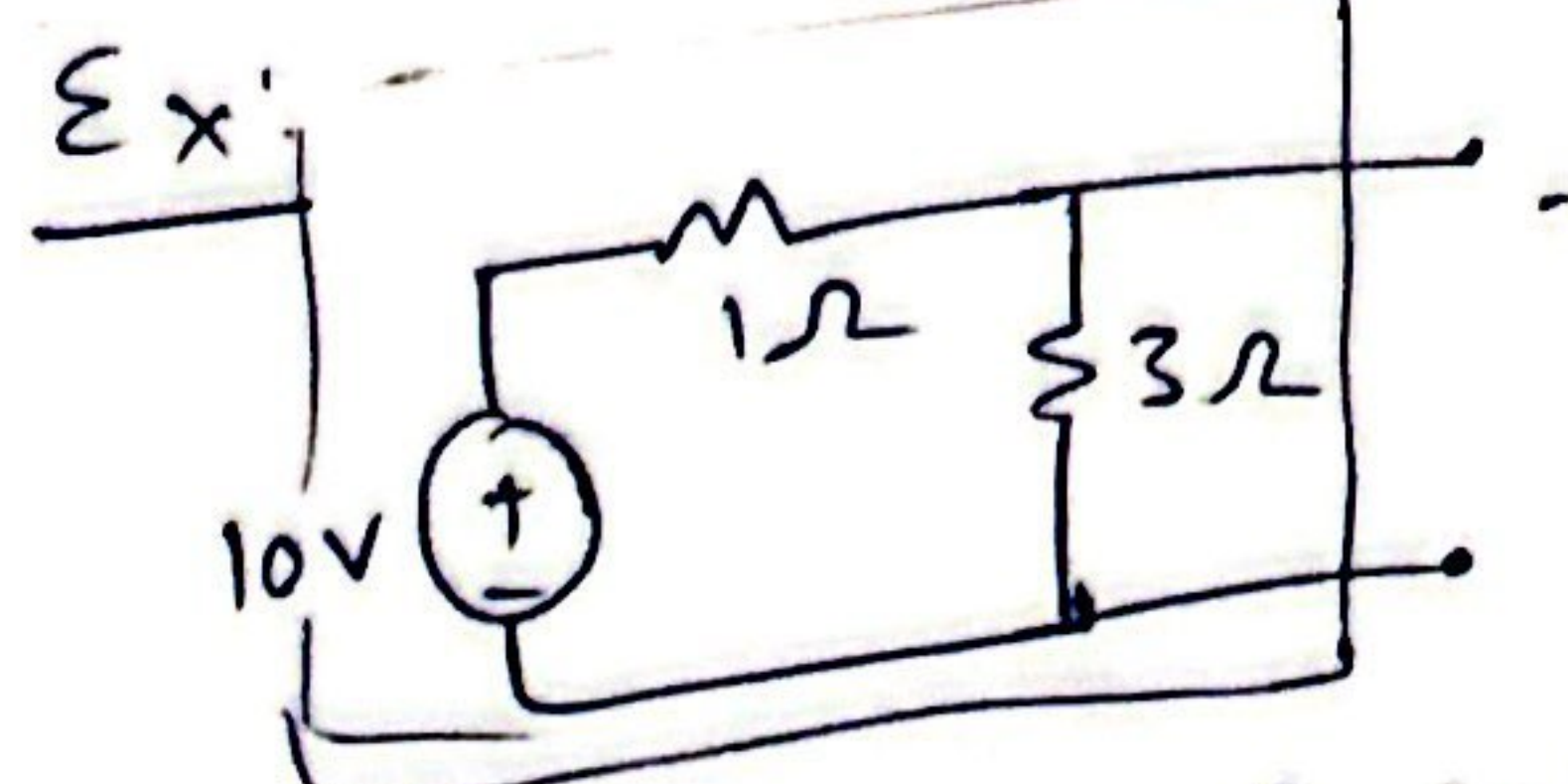
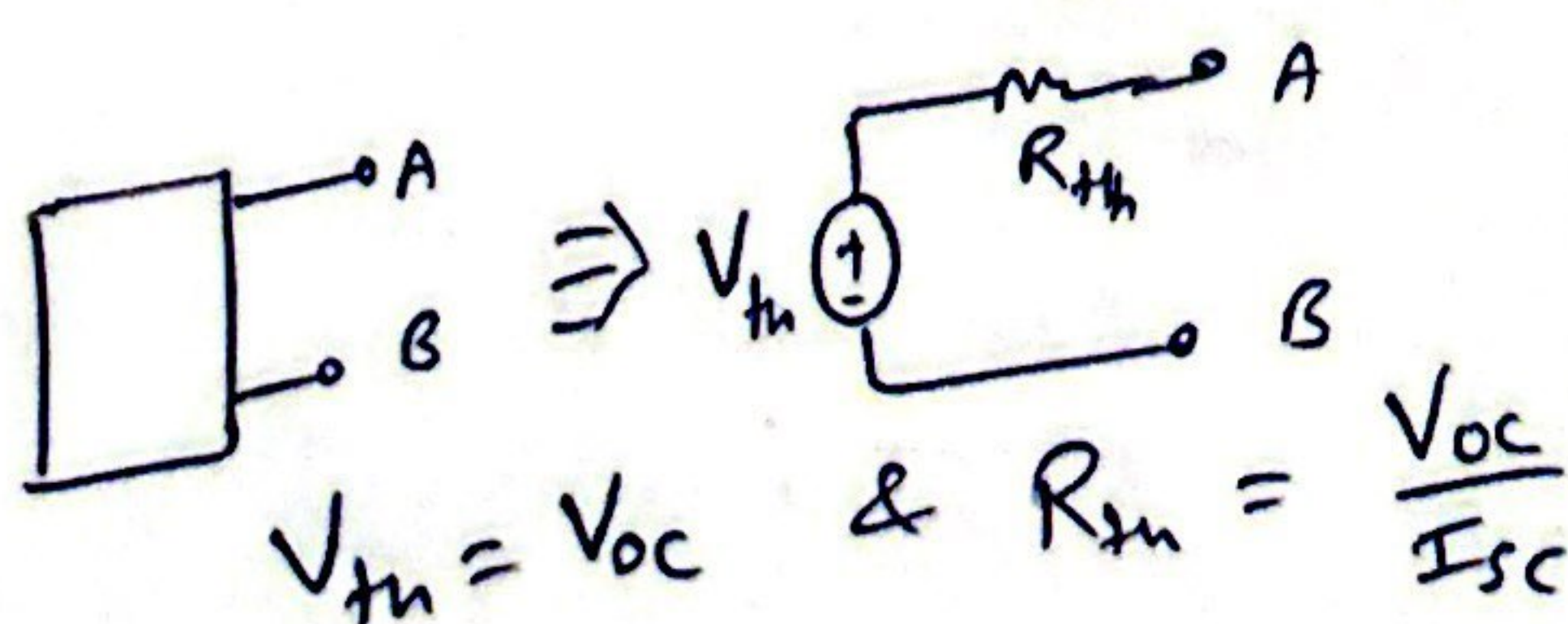
1) find open ckt voltage V_{oc}



2) find short ckt current I_{sc}



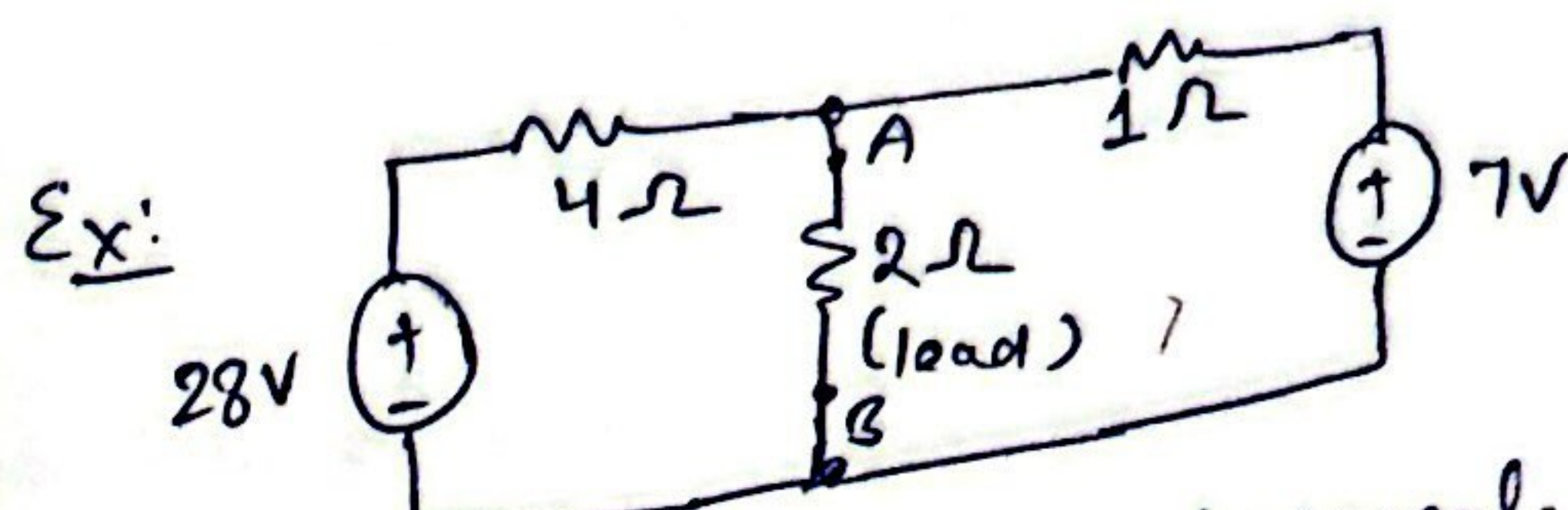
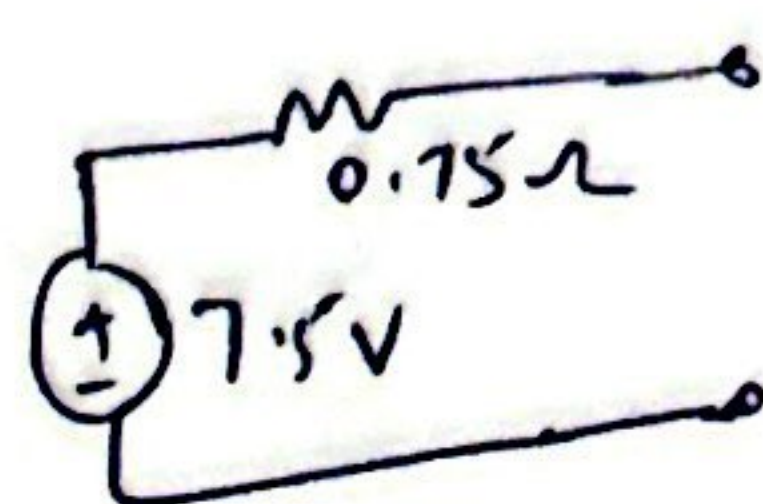
3) $V_{th} = V_{oc}$ & $R_{th} = \frac{V_{oc}}{I_{sc}}$



find Thevenin Equivalent

$$V_{oc} = 7.5 \text{ V}, I_{sc} = 10 \text{ A}$$

$$R_{th} = 0.75 \Omega$$



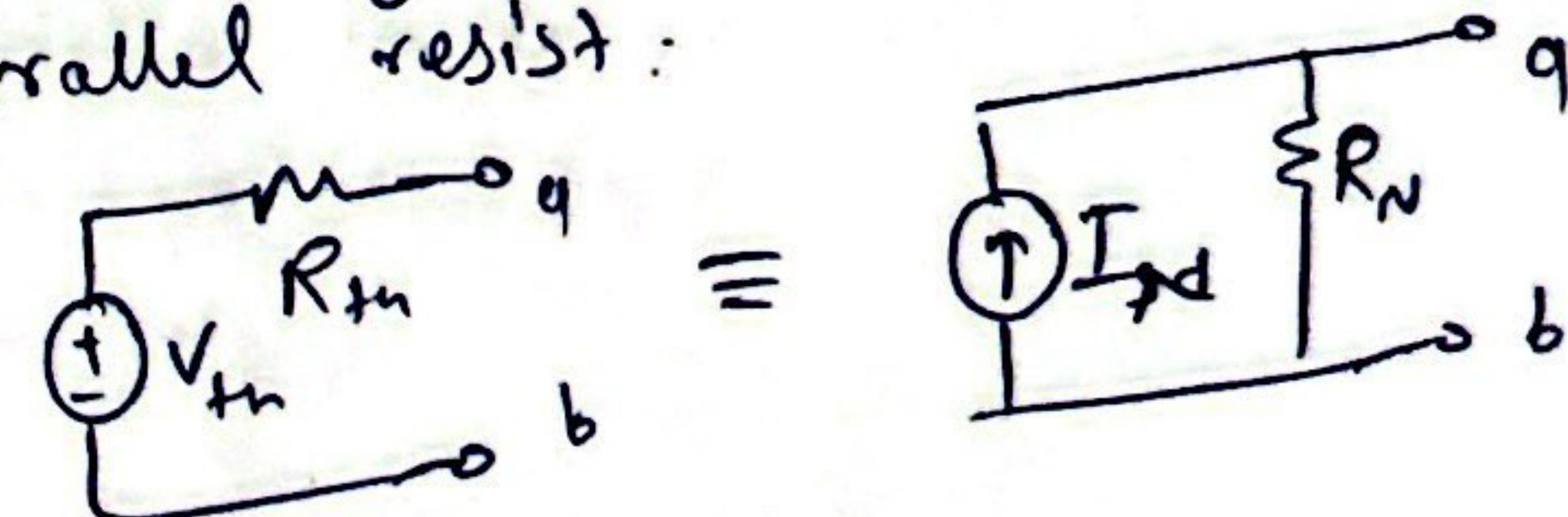
find thevenin across terminal A-B

$$I_{sc} = 14 \text{ A}; V_{oc} = 11.2$$

$$R_{th} = 0.8 \Omega$$

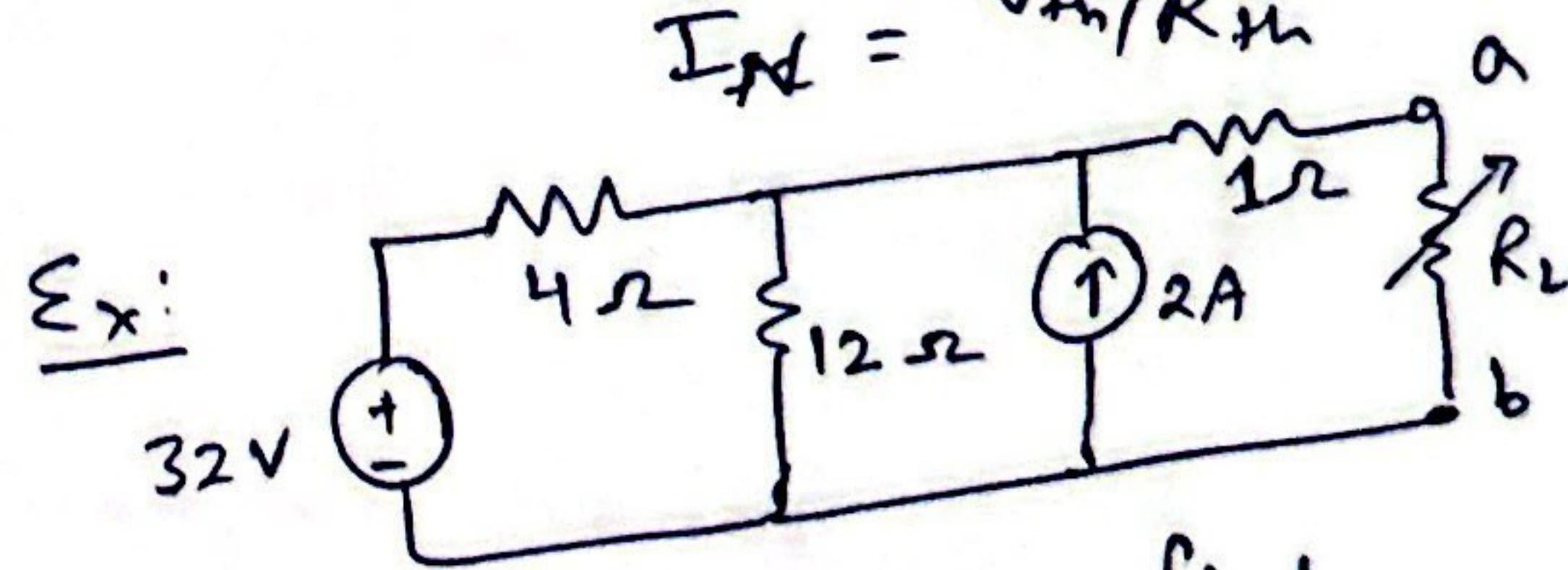
Norton Eqⁿ Ckt:

Just like thevenin, a ckt can be replaced by a current source & parallel resist:



where $R_N = R_{th}$

$$I_{N} = \frac{V_{th}}{R_{th}}$$



Find th. across a-b. find i through $R_L = 6 \Omega$ & 36Ω

Ans: $V_{oc} = 30 \text{ V}, I_{sc} = 7.5$

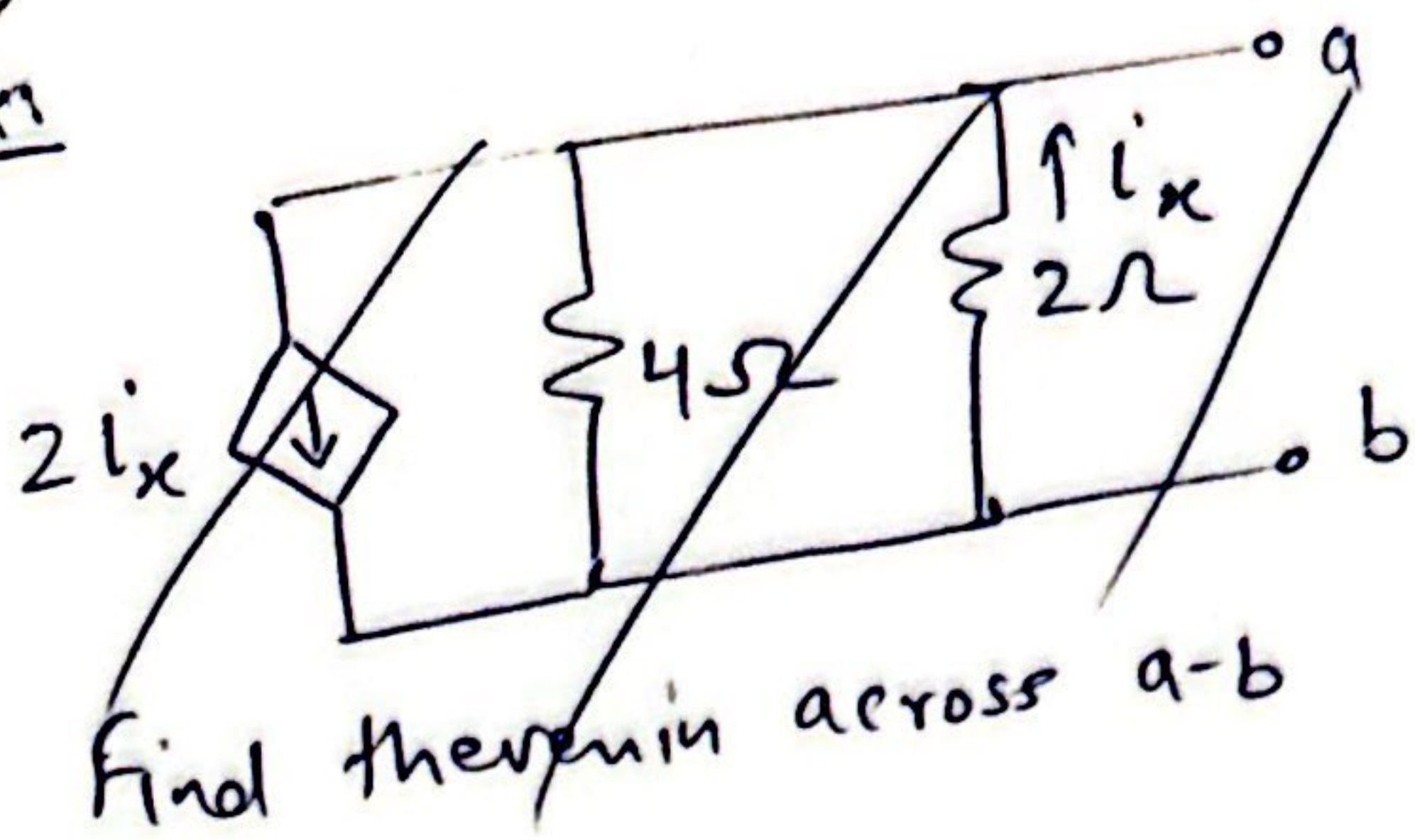
$$R_{th} = 4 \Omega$$

a) $R_L = 6 \Omega, i_L = 3 \text{ A}$

b) $R_L = 36 \Omega, i_L = 0.75 \text{ A}$

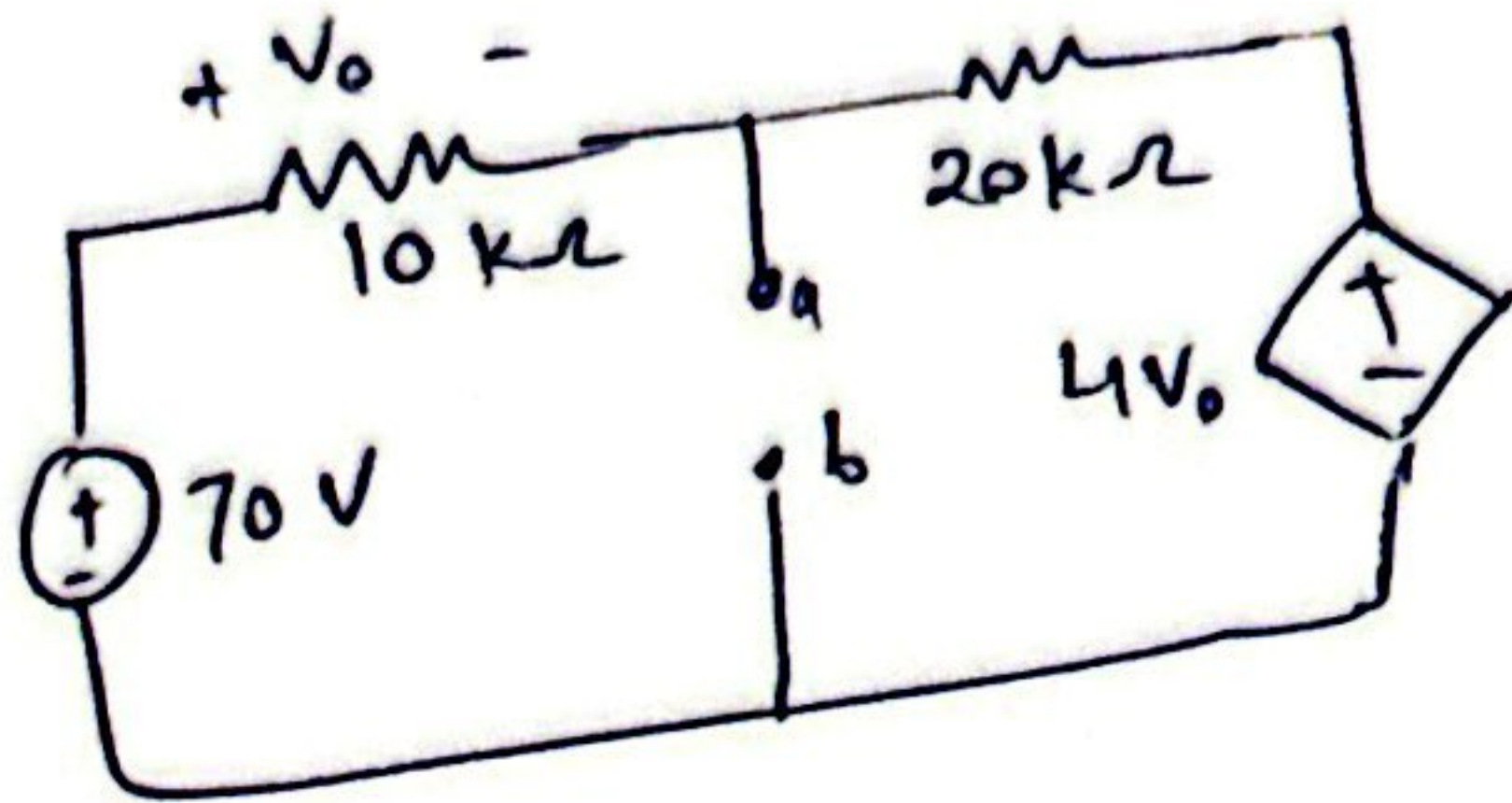
Another way to calculate R_{th} with independent sources only by turning off

6
Ex:



Soln:

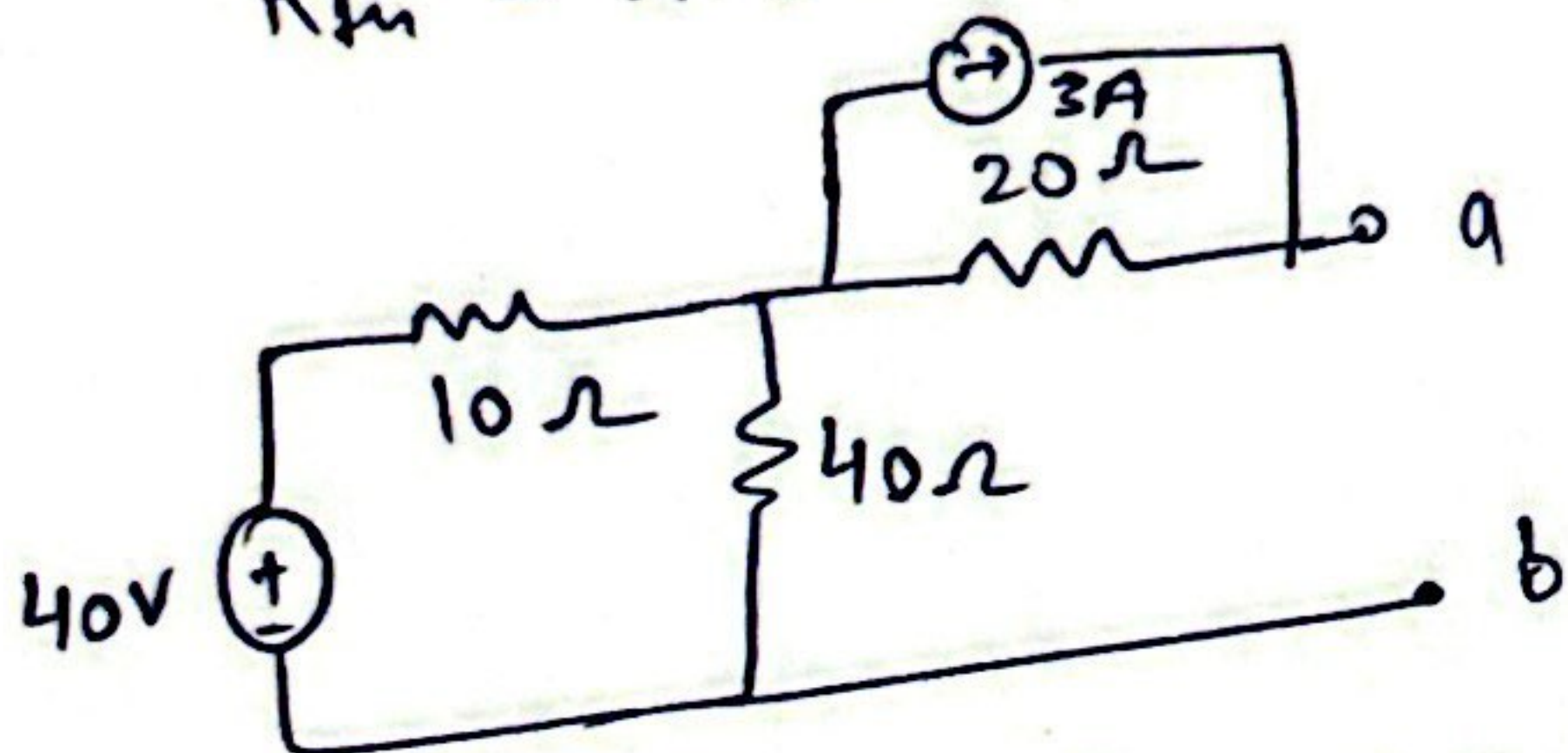
Ex:



find V_{th} .

Soln: $V_{th} = 60V$, $I_{sc} = 21mA$
 $R_{th} = 2.857 k\Omega$

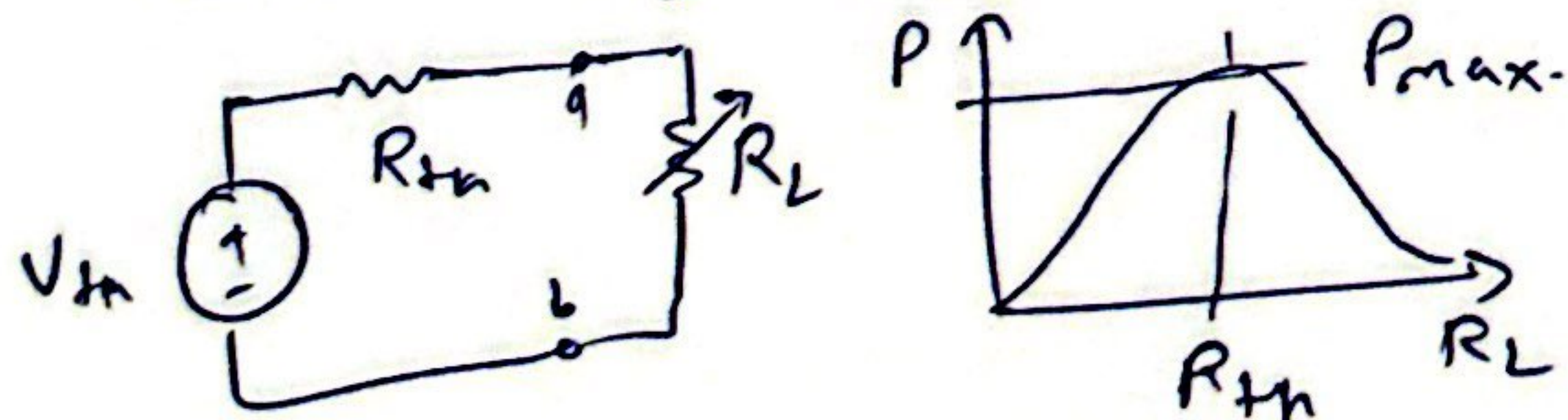
Ex:



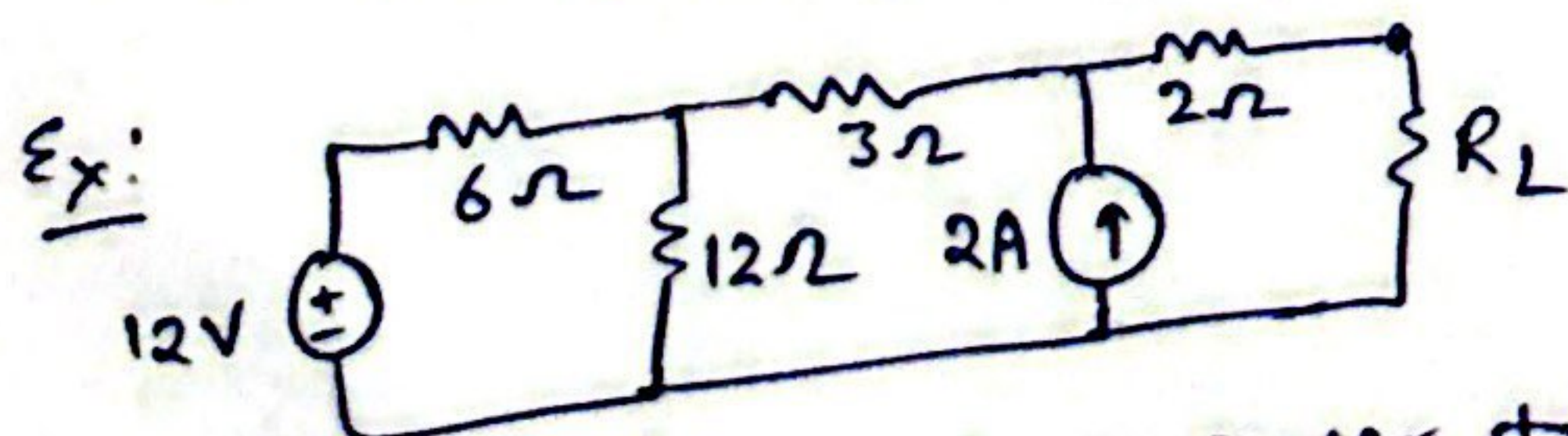
Soln: $V_{th} = 92V$, $R_{th} = 28\Omega$

Maximum Power Transfer

max. power is transferred to the load from a network when load resistance equals the Thevenin resistance seen from the load. $R_L = R_{th}$



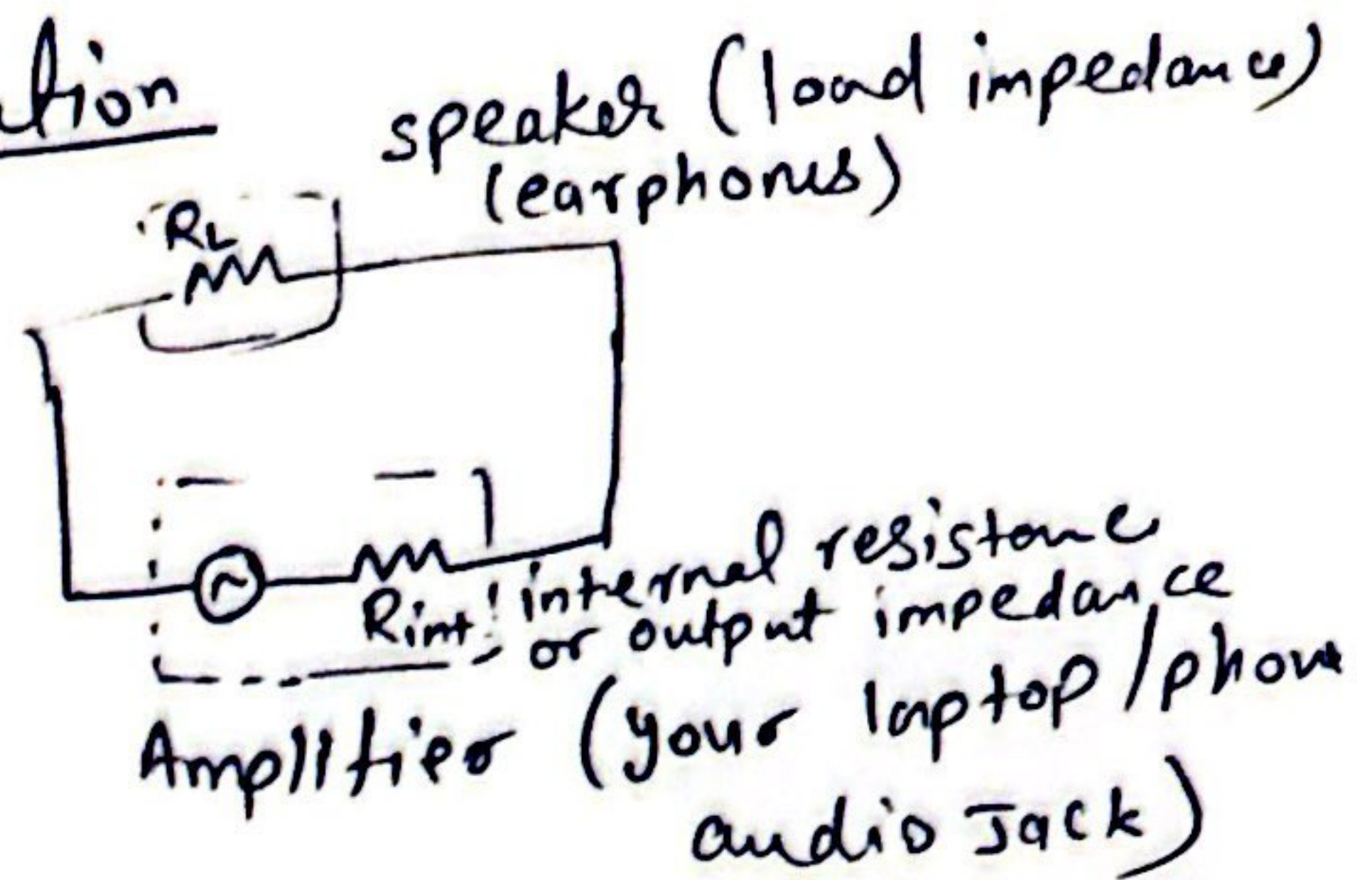
⇒ Proof: students can prove.



find R_L for maximum power transfer

Soln: $R_L = 9\Omega$, $P_{max} = 13.44 W$

Application



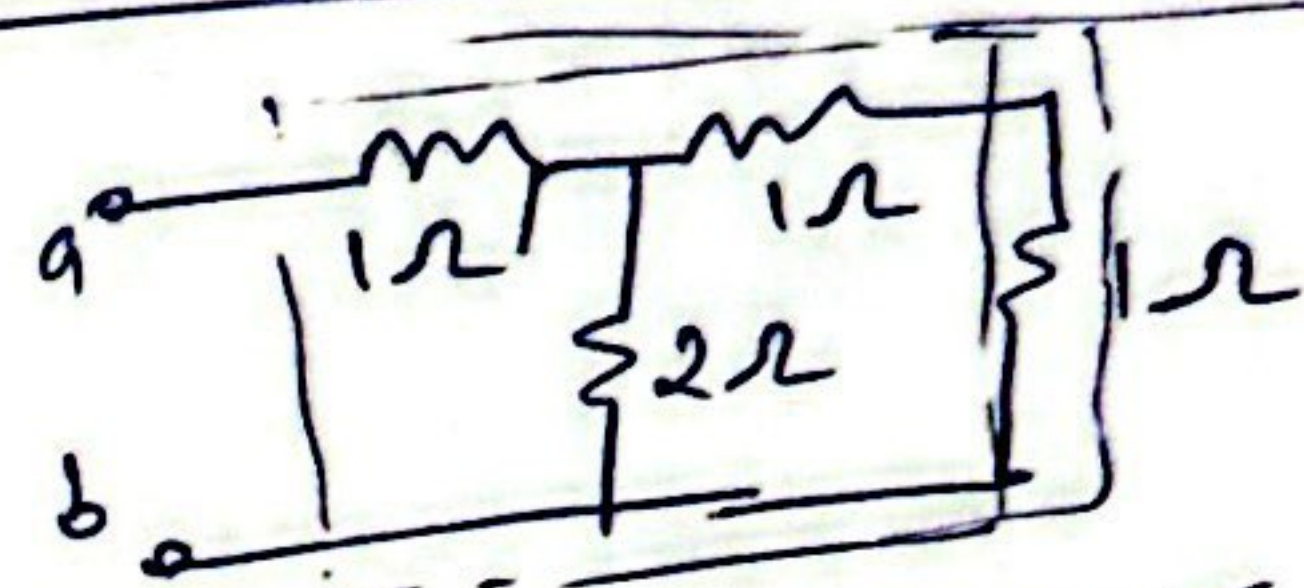
for laptop: typical output impedance
 $R_{int} \approx 20\Omega$

best to use 20Ω speakers
 ⇒ that's why high impedance ($\sim 600\Omega$) speakers are high quality but can't operate with your phone

Some basic circuit concepts

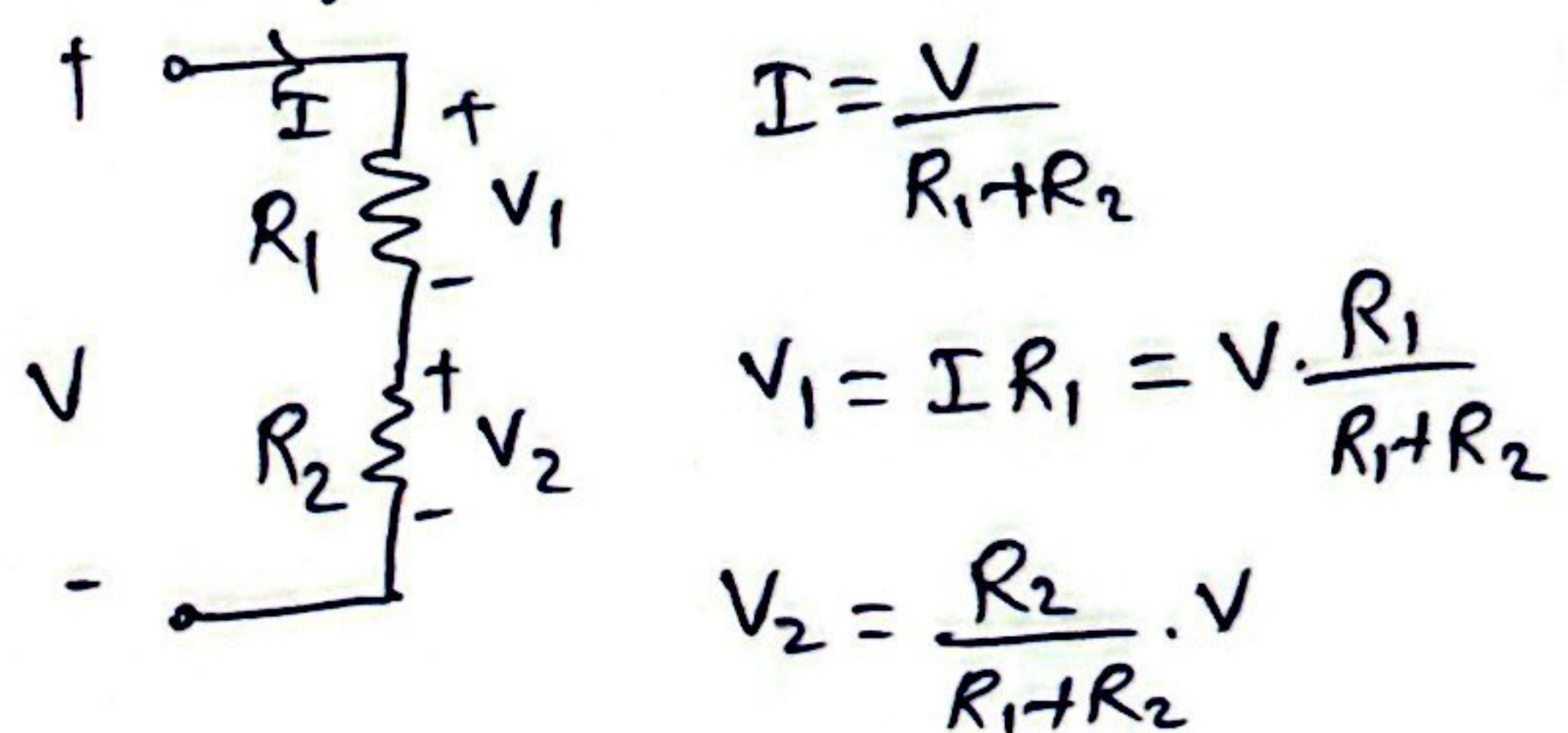
Series & parallel resistances

Ex:



Eqⁿ resistance a-b: (a) 0.5 (c) 2
 (b) 1 (d) 5

⇒ Resistor in series act as voltage divider



⇒ Current divider: resistor in parallel act as

